

CRM Application

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Overview

The **CRM** (Customer Relationship Management) system is designed to optimize customer interactions, manage sales data, and track inventory efficiently. Built using **Spring Boot** for the backend, the system enables businesses to leverage their customer data for informed decision-making and enhanced customer service. This report focuses on the backend architecture, key features, and core operations of the CRM system.

CRM System Features

- **Customer Management:** Enables the creation, updating, and deletion of customer records.
- **Sales Tracking:** Allows the recording of sales transactions and retrieval of sales data for analysis.
- **Inventory Management:** Manages stores and their stock levels. Items can be added, updated, or removed from the inventory.

Backend (Spring Boot)

Entities and Database Modeling

The CRM system relies on several core entities, each mapped to MySQL database tables through **Spring Data JPA**. These entities represent essential business objects such as customers, sales, and inventory, forming the

backbone of the system's data model. The system utilizes JPA annotations like `@Entity`, `@Id`, and `@OneToMany` to define entity relationships and structure.

Service Layer

The **Service Layer**, often found in the *models* package, is a critical component of the backend architecture, encapsulating the business logic of the CRM system. It ensures data consistency and the efficient management of customers, sales, and inventory. The service layer also performs necessary validations to ensure data integrity before executing operations, ensuring that the system operates smoothly and securely.

API Controllers

The backend provides a set of RESTful API endpoints that expose the CRM system's core functionalities to the frontend. These endpoints are implemented using **Spring MVC** annotations. The API controllers handle requests related to app data.

Frontend and Communication with Backend

The frontend of the CRM system, typically built with **Vue.js**, interacts with the backend through REST API calls. The frontend uses **axios** to send relevant requests enabling real-time data updates.

Conclusion

The CRM system provides features for managing customers, sales, and inventory, serving as a critical backbone for data analysis applications.